PLANT MAINTENANCE & OPERATIONAL MANUAL

STANDARD OPERATING PROCEDURE — STANDARD INDUSTRIAL MACHINERY

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SECTION 1 — GENERAL MAINTENANCE PRINCIPLES

Industrial equipment installed across processing facilities must undergo periodic inspection and preventative servicing in accordance with scheduled operational intervals. All technicians are required to document sensor variations, vibration signatures, and thermal fluctuations prior to performing physical disassembly.

SECTION 7 — PUMP OVERHEATING & DIAGNOSTIC PROCEDURES

Repeated high-temperature operation in heavy-duty centrifugal and industrial water pumps poses significant risks of mechanical fatigue, seal degradation, and catastrophic motor burnout. Continuous thermal monitoring is required for all Class-P units.

7.1 Diagnostic Triggers & Operating Limits

Normal operational temperatures for primary pump housings range between 60°C and 75°C. Any sustained thermal reading exceeding 85°C indicates abnormal friction or fluid resistance, requiring immediate technical evaluation.

7.2 Primary Causes of Thermal Elevation

When elevated operating temperatures are recorded across multiple sensor points, maintenance teams must investigate four primary root causes:

1. **Bearing wear:** Internal degradation or severe fatigue of rolling element bearings causing mechanical friction.
2. **Insufficient lubrication:** Inadequate or degraded lubricant film leading to high friction coefficients inside bearing housings.
3. **Cooling system blockage:** Obstructed coolant jackets, fouled radiator fins, or restricted recirculation paths preventing heat dissipation.
4. **Excessive operating load:** Operating the pump above nominal rated hydraulic capacity or against excessive head pressure.

CRITICAL DIAGNOSTIC CORRELATION

If abnormal vibration (high-frequency chatter or axial displacement) is observed together with elevated operating temperature, bearing condition should be inspected immediately as the primary suspect cause. Friction caused by deteriorated bearing races directly induces high thermal output combined with mechanical vibration.

7.3 Action Matrix & Troubleshooting Protocol

Observed Symptom Combination	Primary Suspect Area	Mandatory Inspection Step
High Temp (>85°C) + Normal Vibration	Cooling system / Lubrication	Inspect fluid levels, flush coolant lines, check grease viscosity.

Observed Symptom Combination	Primary Suspect Area	Mandatory Inspection Step
High Temp (>85°C) + Abnormal Vibration	Bearing Assembly / Shaft Alignment	Inspect bearing race degradation, measure axial play, check alignment.
High Temp + Cavitation Noise	Suction Line / Operating Load	Check inlet valve positioning, clear suction strainer, check load.

SAFETY PROCEDURE — MANDATORY SHUTDOWN PROTOCOL

The equipment must be safely shut down and locked out (LOTO protocol) before any maintenance inspection or physical dismantling begins. Do not continue operation if temperature continues to increase beyond the permitted operating range (90°C threshold).